

Foundations of machine learning
Online convex optimization

Maximilian Kasy

Department of Economics, University of Oxford

Winter 2025

Outline

- Setup of online convex optimization:
 - Iteratively choose $\mathbf{x}_t$.
 - Observe loss $f_t(\mathbf{x}_t)$ and gradient $\nabla f_t(\mathbf{x}_t)$.
- Baseline algorithm:
Online gradient descent (OGD).
- Adversarial regret guarantee for OGD.
- Connection to related settings:
 - Adversarial online learning.
 - Stochastic gradient descent.
 - Multiarmed bandits.

Takeaways for this part of class

- Online convex optimization provides a natural framework to connect learning theory and optimization theory.
- Adversarial regret guarantee:
Adversarial regret grows at a rate of $\sqrt{T}$.
- Other settings can be reduced to online convex optimization:
 - Stochastic gradient descent:
Adversarial bounds imply stochastic bounds.
Return average of $\mathbf{x}_t$ at the end.
 - Bandit settings:
Form unbiased estimators of loss and gradients using inverse probability weighting.

Online gradient descent

Connection to other learning problems

References

Setup

- Sequential choices $\mathbf{x}_t \in \mathcal{X}$, where $\mathcal{X}$ is convex.
- Convex loss functions $\mathbf{f}_t(\cdot)$.
- Observable, after choice of $\mathbf{x}_t$:
 - Cost $\mathbf{f}_t(\mathbf{x}_t)$.
 - Gradient $\nabla \mathbf{f}_t(\mathbf{x}_t)$.
- Regret:

$$R_T = \sum_{t=1}^T f_t(\mathbf{x}_t) - \sum_{t=1}^T f_t(\mathbf{x}^*),$$

where

$$\mathbf{x}^* = \operatorname{argmin}_{\mathbf{x} \in \mathcal{X}} \sum_{t=1}^T f_t(\mathbf{x}).$$

Online gradient descent

- For each $t = 1$ to T :
 1. Play $\mathbf{x}_t$.
 2. Observe $\nabla_t = \nabla f_t(\mathbf{x}_t)$.
 3. Update with a gradient step:

$$\mathbf{y}_{t+1} = \mathbf{x}_t - \eta_t \cdot \nabla_t.$$

4. Project into $\mathcal{K}$:

$$\mathbf{x}_{t+1} = \Pi_{\mathcal{K}} \mathbf{y}_{t+1}.$$

- The stepsizes η_t are tuning parameters, to be specified.

Adversarial regret bound

Theorem

- Consider online gradient descent with step-sizes

$$\eta_t = \frac{D}{G\sqrt{t}},$$

where

$$\|x - y\| \leq D \quad \forall x, y \in \mathcal{K}, \quad \|\nabla f(x)\| \leq G \quad \forall x \in \mathcal{K}.$$

- Then:

$$R_T \leq \frac{3}{2}GD\sqrt{T}.$$

Proof

- By convexity of f_t :

$$f_t(x_t) - f_t(x^*) \leq \nabla_t \cdot (x_t - x^*).$$

- By orthogonal projection:

$$\|x_{t+1} - x^*\| \leq \|y_{t+1} - x^*\|.$$

- By definition of gradient update:

$$\|y_{t+1} - x^*\|^2 = \|x_t - x^*\|^2 + \eta_t^2 \|\nabla_t\|^2 - 2\eta_t \nabla_t \cdot (x_t - x^*).$$

- Rearrange. By upper bound on ∇_t :

$$2\nabla_t \cdot (x_t - x^*) \leq \frac{\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2}{\eta_t} + \eta_t G^2.$$

Proof continued

Collect bounds and sum across t :

$$\begin{aligned}2R_t &\leq 2 \sum_t \nabla_t \cdot (x_t - x^*) \\&\leq \sum_t \left[\frac{\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2}{\eta_t} + \eta_t G^2 \right] \\&\leq \sum_t \|x_t - x^*\|^2 \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}} \right) + G^2 \cdot \sum_t \eta_t && (1/\eta_0 = 0, \|x_{T+1} - x^*\|^2 \geq 0) \\&\leq D^2 \frac{1}{\eta_T} + G^2 \cdot \sum_t \eta_t && (\text{telescoping series}) \\&\leq 3DG\sqrt{T} && (\text{definition of } \eta_t; \sum_{t=1}^T 1/\sqrt{t} \leq 2\sqrt{T}).\end{aligned}$$

□

Online gradient descent

Connection to other learning problems

References

Online learning

- Recall the online learning problem:
 - Expert predictions $\hat{Y}_{h,t}$.
 - Loss $L(\hat{Y}_t, Y_t)$.
- Map into online convex optimization:
 - Weight vector $\mathbf{x}_t = (\mathbf{x}_{h,t})$ in the simplex $\mathcal{K}$.
 - Prediction:

$$\hat{Y}_t = \sum_h \mathbf{x}_{h,t} \cdot \hat{Y}_{h,t}.$$

- Gradient:

$$\nabla_t = \left(\hat{Y}_{h,t} \right)_h \cdot \partial_{\hat{Y}} L(\hat{Y}_t, Y_t).$$

Stochastic gradient descent

- Recall the stochastic optimization setting:
 - Our goal is to minimize $f(\mathbf{x})$ w.r.t. $\mathbf{x}$.
 - We observe unbiased gradient estimates ∇_t :

$$E[\nabla_t | \mathbf{x}_t] = \nabla f(\mathbf{x}_t).$$

Think: $\nabla_t = \nabla m(\mathbf{x}, \mathbf{Z}_t)$.

- Stochastic gradient descent:
 1. Apply online gradient descent.
 2. Return $\bar{\mathbf{x}}_T = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t$.

Regret bound for stochastic gradient descent

Assume $E[\|\nabla_t\|^2] \leq G^2$. Then

$$E[f(\bar{x}_T)] - f(x^*) \leq \frac{3GD}{2\sqrt{T}}.$$

Sketch of proof:

$$\begin{aligned} E[f(\bar{x}_T)] - f(x^*) &\leq \frac{1}{T} \sum_{t=1}^T E[f(x_t) - f(x_*)] && \text{(convexity)} \\ &\leq \frac{1}{T} \sum_{t=1}^T E[\nabla_t \cdot (x_t - x_*)] && (E[\nabla_t | x_t] = \nabla f(x_t)) \\ &\leq \frac{3GD}{2\sqrt{T}}. && \text{(Theorem for OGD)} \end{aligned}$$

□

Multi-armed bandits

- Coming up next in class.
- Only observe loss L_t for actions actually chosen.
- For randomized algorithms, we can form unbiased estimators of the gradient of reward:

$$\nabla_t = \left(L_t \cdot \frac{\mathbf{1}(D_t = d)}{x_{d,t}} \right)_d$$

- This allows us to reduce the adversarial bandit problem to an online convex optimization problem.

References

Hazan, E. (2016). Introduction to online convex optimization. Foundations and Trends in Optimization, 2(3-4):157–325, chapter 3.